NextGen
OLYMPIAD FOUNDATION

GRADE 6

STEM OLYMPIAD

National Level Examination

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

Sample paper

SECTION A – MATHEMATICS & QUANTITATIVE REASONING

Q1. What is the value of (-15) +(-8) -(-12)+20?
A. 7
B. 8
C. 9
D. 11

Q2. What is 3/4 + 5/8?
A. 7/8
B. 9/8
C. 11/8
D. 13/8

Q3. What is 2.5 x 0.4?
A. 0.1
B. 1.0
C. 10
D. 100

Q4. The ratio 18:24 in simplest form is:
A. 2:3
B. 3:4
C. 4:5
D. 6:8

Q5. If 3x - 5 = 2x+7, what is the value of 2x + 1?
A. 12
B. 24
C. 25
D. 26

Q6. A rectangle has a perimeter of 38cm. If its length is 12 cm, What is the area of the rectangle?
A. 84 cm^2
B. 96 cm^2
C. 60 cm^2
D. 42 cm^2

Q7. A cuboid measures 5 cm x 4 cm x 3 cm. What is its volume?
A. 12 cm^3
B. 20 cm^3
C. 47 cm^3
D. 60 cm^3

Q8. The mean of 6, 8, 10, 12 and 14 is:
A. 9
B. 10
C. 11
D. 12

Q9. Which number is prime?
A. 21
B. 29
C. 33
D. 39

Q10. The LCM of 8 and 12 is:
A. 16
B. 20
C. 24
D. 48

Q11. If 5 notebooks cost Rs. 210, what do 8 notebooks cost at the same rate?
A. Rs 288
B. Rs 320
C. Rs 336
D. Rs 360

Q12. A line segment has:
A. one endpoint
B. two endpoints
C. no endpoints
D. three endpoints

Q13. An angle of 270° is:
A. acute
B. obtuse
C. reflex
D. straight

Q14. A car travels 180 km in 3 h at constant speed. Its speed is:
A. 50 km/h
B. 60 km/h
C. 70 km/h
D. 90 km/h

Q15. What is 25% of 240?
A. 40
B. 50
C. 60
D. 80

Sample paper

SECTION B – SCIENCE & SCIENTIFIC REASONING

Q16. Which nutrient is mainly needed for body growth and repair?
A. Protein
B. Water only
C. Roughage only
D. Mineral salt only

Q17. Which plant tissue carries water upward from roots?
A. Xylem
B. Phloem
C. Petal
D. Stomata

Q18. Which method separates an insoluble solid from a liquid?
A. Filtration
B. Evaporation only
C. Magnetism only
D. Freezing

Q19. Which joint allows movement in many directions, as in the shoulder?
A. Hinge
B. Ball-and-socket
C. Fixed
D. Pivot only

Q20. Which change is chemical?
A. Melting wax
B. Dissolving sugar
C. Burning paper
D. Freezing water

Q21. The standard unit of temperature is ?
A. Fahrenheit
B. kelvin
C. °Celsius
D. Joule

Q22. A shadow is shortest around midday mainly because the Sun is:
A. higher in the sky
B. below the horizon
C. blocked by Earth
D. producing less light

Q23. A simple electric circuit works when it provides:
A. a complete conducting path
B. only one wire touching the bulb
C. an open switch
D. no cell

Q24. Which end of a freely suspended magnet points towards the geographical North pole?
A. North-seeking pole
B. South-seeking pole
C. Magnetism heats it
D. The spoon creates heat

Q25. Why does a metal spoon in hot tea become warm?
A. Heat is conducted through the metal
B. Sound moves through it
C. Magnetism heats it
D. The spoon creates heat

Q26. Which gas is most abundant in air?
A. Oxygen
B. Nitrogen
C. Carbon dioxide
D. Water vapour

Q27. Which process returns water vapour to liquid droplets in clouds?
A. Condensation
B. Evaporation
C. Sublimation
D. Melting

Q28. Which adaptation best helps a fish move through water?
A. Streamlined body
B. Broad flat feet
C. Dry scales only
D. Long eyelashes

Q29. Four different waste items were buried in moist soil under identical conditions. After six weeks, which items will show the Highest rate of decomposition without leaving residue?
A. A laminated paper
B. Fruits peels and leftover cooked rice
C. A thin, single-use plastic bag
D. An aluminum soft drink can

Q30. Which force slows a sliding book on a table?
A. Friction
B. Magnetism only
C. Light
D. Sound

Sample paper

SECTION C – ENGINEERING, TECHNOLOGY & DATA

Q31. A bridge truss uses triangles because triangles are:
A. rigid and resist shape change
B. always lighter than air
C. magnetic
D. transparent

Q32. A student compares evaporation from two dishes. Which condition should differ in a fair test of surface area?
A. Only the exposed surface area
B. Temperature of water
C. Amount of water
D. Sunlight exposure

Q33. A lever has the load 20 cm from the fulcrum and effort 60 cm from the fulcrum. Ignoring losses, the effort needed is:
A. greater than the load
B. less than load
C. One-third of the load
D. zero

Q34. Which circuit arrangement keeps other bulbs glowing if one bulb fails?
A. Parallel
B. Series only
C. Open circuit
D. No wires

Q36. A sensor records 20°C, 22°C, 24°C, 26°C. What is the rate of change per reading?
A. 1°C
B. 2°C
C. 3°C
D. 4°C

Q37. A robot moves 5 m East, 3 m North, 2 m West. Its final position from the start is:
A. 3 m East and 3 m North
B. 7 m East and 3 m North
C. 3 m West and 3 m North
D. 5 m East and 1 m North

Q38. A paper airplane travels farther after its wings are adjusted. What should be done to test whether the wing change caused the improvement?
A. Repeat trials while keeping throw method the same
B. Use a different thrower each time
C. Change wing shape and throw strength together
D. Use only one flight

Q39. A 500 W device runs for 2 h. How much energy does it use in kilowatt-hours?
A. 0.5 kWh
B. 1 kWh
C. 2 kWh
D. 1000 kWh

Q35. A 2 kg object has greater mass than a 500 g object by:

A. 0.5 kg
B. 1.0 kg
C. 1.5 kg
D. 2.5 kg

Q40. A water tank holds 150 L. It is 3/5 full. How much water is in it?

A. 60 L
B. 75 L
C. 90 L
D. 120 L

SECTION D – STEM OLYMPIAD CHALLENGE

Q41. If a 12-tooth gear drives a 36-tooth gear, the larger gear makes one turn when the smaller makes:

A. 1 turn
B. 2 turns
C. 3 turns
D. 6 turns

Q42. Which graph is best for showing how temperature changes over time?

A. Line graph
B. Pie chart only
C. Pictograph only
D. Venn diagram

Q43. A solar cell changes light energy mainly into:

A. electrical energy
B. sound energy
C. chemical energy only
D. gravitational energy

Q44. Which design most reduces heat transfer from a hot bottle?

A. Vacuum layer with insulating stopper
B. Bare metal wall
C. Wide open top
D. Thin copper covering

Q45. A machine lifts a 100 N load using 50 N effort. Its mechanical advantage is:

A. 0.5
B. 1
C. 2
D. 50

Q46. A seed experiment uses 20 seeds in light and 20 in dark, both with equal water and temperature. What is the independent variable?

A. Light condition
B. Number of seeds
C. Water amount
D. Seed type

Q47. A cyclist covers 6 km in 20 min. At the same speed, how far in 1 h?

A. 12 km
B. 15 km
C. 18 km
D. 24 km

Q48. Which observation most strongly suggests a new substance formed?

A. Gas forms and colour changes during a reaction
B. Ice melts
C. Salt dissolves
D. Water freezes

Q49. A 10 m ramp rises 2 m vertically. Compared with lifting straight up, the ramp generally requires:

A. less force over more distance
B. more force over less distance
C. no force
D. the same force and no extra distance

Q50. A STEM team tests three tower designs. The best evidence for choosing one is:

A. repeatable load-test data
B. the tallest student's opinion
C. the prettiest colour
D. one lucky trial

Sample Paper